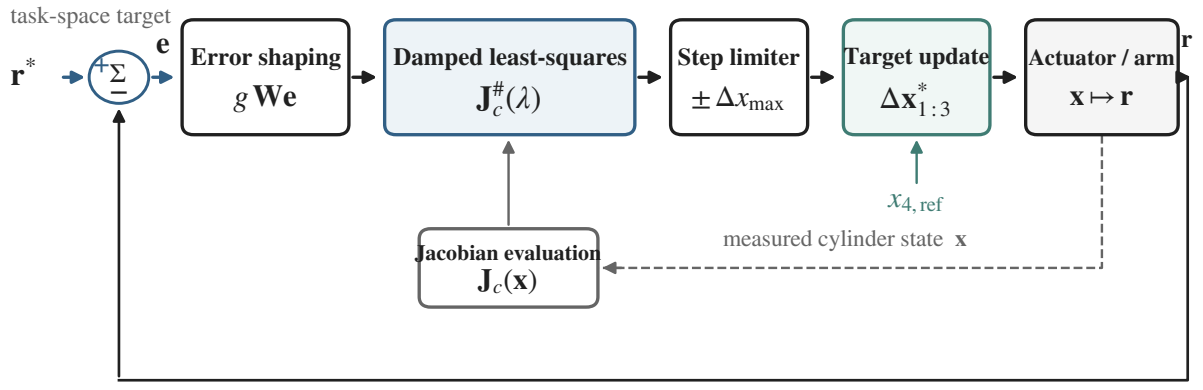
task-space target
$\mathbf{r}^*$
$+\Sigma-$
$\mathbf{e}$
Error shaping
$g\,\mathbf{We}$
Damped least-squares
$\mathbf{J}_c^{\#}(\lambda)$
Step limiter
$\pm\Delta x_{\max}$
Target update
$\Delta\mathbf{x}^*_{1:3}$
Actuator / arm
$\mathbf{x}\mapsto\mathbf{r}$
$\mathbf{r}$
$x_{4,\,\mathrm{ref}}$
measured cylinder state $\mathbf{x}$
Jacobian evaluation
$\mathbf{J}_c(\mathbf{x})$
measured cutter position $\mathbf{r}$